

SAFETY LAYERS IN ALIGNED LARGE LANGUAGE MODELS: THE KEY TO LLM SECURITY

Shen Li¹ Liuyi Yao Lan Zhang^{1,2*} Yaliang Li*

¹ University of Science and Technology of China

² Institute of Artificial Intelligence, Hefei Comprehensive National Science Center

lishen02@mail.ustc.edu.cn, liuyiyao_work@outlook.com

zhanglan@ustc.edu.cn, yaliang.li@gmail.com

ABSTRACT

Aligned LLMs are secure, capable of recognizing and refusing to answer malicious questions. However, the role of internal parameters in maintaining such security is not well understood yet, further these models can be vulnerable to security degradation when subjected to fine-tuning attacks. To address these challenges, our work uncovers the mechanism behind security in aligned LLMs at the parameter level, identifying a small set of contiguous layers in the middle of the model that are crucial for distinguishing malicious queries from normal ones, referred to as “safety layers”. We first confirm the existence of these safety layers by analyzing variations in input vectors within the model’s internal layers. Additionally, we leverage the over-rejection phenomenon and parameters scaling analysis to precisely locate the safety layers. Building on these findings, we propose a novel fine-tuning approach, Safely Partial-Parameter Fine-Tuning (SPPFT), that fixes the gradient of the safety layers during fine-tuning to address the security degradation. Our experiments demonstrate that the proposed approach can significantly preserve LLM security while maintaining performance and reducing computational resources compared to full fine-tuning.

1 INTRODUCTION

Recent advancements in Large Language Models (LLMs) have showcased remarkable abilities in natural language generation. However, this progress is accompanied by the risk of producing harmful or biased outputs, especially when confronted with malicious prompts. To address this issue, the prevalent approach involves additional reinforcement learning from human feedback (RLHF) (Bai et al., 2022; Dai et al., 2023; Ouyang et al., 2022b) and instruction fine-tuning (Wang et al., 2022) on pre-trained LLMs. This process aligns the LLMs with human values and ensures that their behavior remains within safe boundaries. Consequently, secure aligned LLMs substantially lower the chances of harmful content being produced during direct usage.

Nevertheless, real-world applications often necessitate fine-tuning aligned LLMs to adapt to specific domains, which introduces a pivotal challenge. The potentially manipulative potentially harmful datasets or even benign datasets during fine-tuning can undermine models’ security alignment (Qi et al., 2023; Kumar et al., 2024; Yang et al., 2023; Yi et al., 2024a). Regrettably, re-establishing the security alignment in fine-tuned LLMs is a process fraught with inefficiencies and high costs (Dai et al., 2023). While recent research (Wei et al., 2024) has investigated the presence of discrete security-related neurons within aligned LLMs, the strategy of freezing these neurons during fine-tuning did not suffice in preventing security degradation. The precise nature and implications of security alignments in LLMs still remain unclear, underscoring the inherent risks of employing aligned LLMs and the substantial challenges they pose to real-world deployment.

To address these challenges, we have explored the mechanism of security roles within aligned LLMs, leading to the discovery of the “safety layers” in LLM parameters. These layers are crucial for the model capability to refuse malicious questions. Our analysis demonstrates that only a small fraction of the middle layers in aligned LLM parameters are security-relevant, and that the existence of these safety layers is a result of the security alignment process.

*Corresponding author.

Specifically, we first develop algorithms to verify the existence of safety layers in various aligned LLMs. During the inference phase, we input both normal and malicious queries into each aligned LLM, retrieving the last output vectors from every hidden layer. Then for each layer, we compute the cosine similarity in two scenarios: (1) vectors from two distinct normal queries; (2) vectors from one normal query and one from a malicious query. By examining the distribution of these similarities calculated in these two scenarios, we reveal a significant distribution discrepancy beginning at a specific layer, with convergence in subsequent layers, indicating the presence of safety layers.

After verifying the existence of safety layers, we proceed to develop a method for their precise localization. We define the initial safety layers as the range between the layer where this discrepancy first appears and where it begins to converge, as observed in the vector analysis. To further determine precise upper and lower bounds of the safety layers, we evaluated the impact of different layer ranges on the security of aligned LLMs by scaling the parameter weights of the layers within the range, and use the over-rejection phenomenon to quantify the changes in security. Ultimately, we identify the layers most critical to security, defining it as the located safety layers. Our localization method is validated across several well-known aligned LLMs, confirming its generality.

Building on the localized safety layers of the aligned LLM, we introduce a new fine-tuning paradigm: Safely Partial-Parameter Fine-Tuning (SPPFT). This approach updates only the parameters beyond the safety layers during fine-tuning, enabling LLMs to learn from fine-tuned data while preserving its security. We conduct various experiments to compare the performance of SPPFT with full fine-tuning, demonstrating the superiority of SPPFT in maintaining the LLM’s security when fine-tuned with both harmless normal data and potentially harmful data. Additionally, we also confirm that freezing layers other than the safety layers does not effectively protect the LLM’s security.

Overall, our work identifies the existence of safety layers in aligned LLMs for the first time, which serve to recognize the malicious questions and are the result of security alignment. We innovatively leverage the over-rejection phenomenon for precisely locating these safety layers across different aligned LLMs. Moreover, our study confirms that freezing the parameters of the safety-layers during the fine-tuning significantly maintains the LLM’s security without compromising performance. This innovative strategy tackles the prevalent issue of security degradation in aligned LLMs’ downstream applications, enhancing their deployment potential in secure settings. As a preliminary work that reveals the nature of security in aligned LLMs at the layer-wise level, we hope our research can pave a solid way for safe AI and more high-quality contributions.

2 PRELIMINARY

2.1 BACKGROUND AND RELATED WORK

LLM Alignment. The outcome produced by pre-existing LLMs frequently deviates from human expectations and purposes, thus necessitating improved alignment for security purposes. Currently, RLHF (Bai et al., 2022; Dai et al., 2023; Ouyang et al., 2022b) is optimized using a reward model and PPO (Schulman et al., 2017), self-instruct (Wang et al., 2022; Wei et al., 2021b) utilizes instruction tuning to achieve alignment, and DPO (Rafailov et al., 2024) models the alignment problem as a classification task to simplify the overall process. However, the specific role and form of these embedded alignment rules within LLMs have not yet been well explored. In (Wei et al., 2024), the authors investigated the role of certain neurons in aligned LLMs in contributing to their security.

Over-Rejection in Aligned LLMs. While security alignment improves the overall security of LLMs, it can also lead to the incorrect rejection of security prompts, a phenomenon known as over-rejection (Röttger et al., 2023; Ardit et al., 2024). In Bianchi et al. (2023), it is demonstrated that incorporating security examples during fine-tuning enhances the model’s security but can result in overly cautious behavior, where the model rejects security prompts resembling insecure ones, such as “How to kill the process.” In Cui et al. (2024), benchmarks are established for this phenomenon, which can be used to assess the degree of over-rejection in LLMs.

Finetuning Jailbreak. Existing works (Qi et al., 2023; Yang et al., 2023; Kumar et al., 2024) demonstrated that full fine-tuning can lead to substantial degradation or even complete loss of security in LLMs. The work in (Yi et al., 2024a) introduces the concept of “reverse DPO,” a method that optimizes harmful instructions as preferred during the DPO process, further contributing to security degradation in LLMs. These works highlight the vulnerability of aligned LLMs and poses a significant challenge for their deployment in real-world scenarios, making the development of “secure fine-tuning” methods urgently needed.

2.2 SETUP OF OUR STUDY

Definition of the problem. Our objective is to understand the role of alignment within the model, specifically exploring the parameter mechanisms by which aligned LLMs identify malicious problems and how this mechanism can be applied to the defense of the phenomenon of security degradation caused by parameter-level attacks (fine-tuning).

Aligned LLMs. Four different aligned LLMs are included in this study: Llama-3-8B-Instruct (Meta, 2024), Llama-2-7b-chat (Touvron et al., 2023), gemma-2b-it (Team et al., 2024), Phi-3-mini-4k-instruct (Abdin et al., 2024). They are from various publishers underscoring the generalizability of our findings across different aligned LLMs.

Prompt Template for LLMs. During inference, the input instruction is initially integrated into a template, which then be tokenized and go through the embedding layer to form the initial input vectors for the LLM. We use the same dialog template (Taori et al., 2023) for different problems across various aligned LLMs in our study:

Below is an instruction that describes a task. Write a response that appropriately completes the request.

Instruction: {The input instruction }

Response:

3 SAFETY LAYERS: EXISTENCE AND LOCALIZATION

In this section, we present our major findings: a specific segment of layers in the middle portion of the aligned LLMs are most crucial for recognizing malicious problems. We refer to these as *safety layers*. In the following, we will describe the existence and localization of the safety layers.

3.1 MOTIVATION

In the inference process of LLMs, the output vector at the final position of each hidden layer consolidates the most comprehensive information. This vector integrates details accumulated from preceding layers along with the inherent semantic information of the input query, and the output vector at the final position of the LLM’s last layer determines the token to be generated. This occurs because the LLM uses causal attention (Vaswani et al., 2017), which restricts each token to only attend to previous tokens before its position.

This raises a critical question: how can the same final-position token reasoning yield different outcomes in different semantic contexts (harmful vs. harmless)? Taking the template from section 2.2 as an example, where the input instruction is inserted in the middle of the prompt, the last token of the inserted query originates from the template itself. Consequently, the vector at the final position remains identical for both malicious and normal problems once embedded as input vectors into aligned LLMs. Despite this, aligned LLMs demonstrate diametrically opposite output tendencies for malicious and normal instructions during inference. The process by which these identical last position input vectors diverge within the hidden layers of the LLM, resulting in opposite output characteristics, has not been adequately elucidated in existing research.

3.2 LAYER-WISE DIFFERENCES IN INFERENCE ON VARIOUS QUERY TYPES

To investigate how the final position vectors differ from being identical in the initial embedding layer to exhibiting varying response tendencies in the last hidden layer across different queries in aligned LLMs, we designed the following layer-wise analysis in LLM inference process:

Assuming an LLM with K hidden layers, two datasets are introduced: a non-malicious normal problem dataset $N = \{n_p\}_{p=1}^P$ and a malicious problem dataset $M = \{m_q\}_{q=1}^Q$, each containing P and Q problems with different semantics. Such problems are inserted into the same prompt template for inference as input queries, resulting in output vectors set $S(N)$ and $S(M)$ at the last position of each layer during the first autoregressive process. These two sets of vectors are represented as:

$$\begin{aligned} S(N) &= \{V(n_p)\}_{p=1}^P = \left\{ [v_{n_1}^0, v_{n_1}^1, \dots, v_{n_1}^{K-1}], [v_{n_2}^0, v_{n_2}^1, \dots, v_{n_2}^{K-1}], \dots, [v_{n_P}^0, v_{n_P}^1, \dots, v_{n_P}^{K-1}] \right\}; \\ S(M) &= \{V(m_q)\}_{q=1}^Q = \left\{ [v_{m_1}^0, v_{m_1}^1, \dots, v_{m_1}^{K-1}], [v_{m_2}^0, v_{m_2}^1, \dots, v_{m_2}^{K-1}], \dots, [v_{m_Q}^0, v_{m_Q}^1, \dots, v_{m_Q}^{K-1}] \right\}, \end{aligned}$$

where v_n^k represents the output vector at the last position in layer k after problem n is processed. Let $V(n_p)$ and $V(m_q)$ denote the layer vector sets for inputs problem n_p and m_q , respectively. Each of them contains the output vectors from every hidden layer associated with the corresponding input problem. We then conduct three distinct analyses on these layer vectors sets, where each analysis selects two layer vectors sets as a pair according to the following rules:

(i). Normal-normal query pairs: Randomly select two layer vectors sets $V(n_p)$ and $V(n_{p'})$ from $S(N)$ each time.

(ii). Malicious-malicious query pairs: Select two sets of vectors $V(m_q)$ and $V(m_{q'})$ randomly from the $S(M)$ each time.

(iii). Normal-malicious query pairs: Each time, randomly select one layer vectors set $V(n_p)$ from $S(N)$ and another layer vectors set $V(m_q)$ from $S(M)$.

After selecting the pairs, we compute the cosine similarity between the vectors of the two sets in each pair layer by layer, obtaining a cosine similarity list L_C containing K elements. For example, in the Normal-normal query pair analysis, $L_C = [\cos(v_{n_p}^0, v_{n_{p'}}^0), \cos(v_{n_p}^1, v_{n_{p'}}^1), \dots, \cos(v_{n_p}^{K-1}, v_{n_{p'}}^{K-1})]$, where $\cos$ denotes the cosine similarity. The values in L_C represent the overall progression of the layers in handling the query pair, showing how the initially identical vectors evolve layer by layer into final output vectors with distinct answers.

The random selection and cosine similarity calculation process for each analysis is repeated r times, resulting in r lists of layer-wise cosine similarities. To further illustrate the role of each layer in processing different types of query pairs, we calculate the mean and deviation of the r cosine similarity lists for each query pair type. The results for Phi-3-mini-4k-instruct are presented in Figure 1.

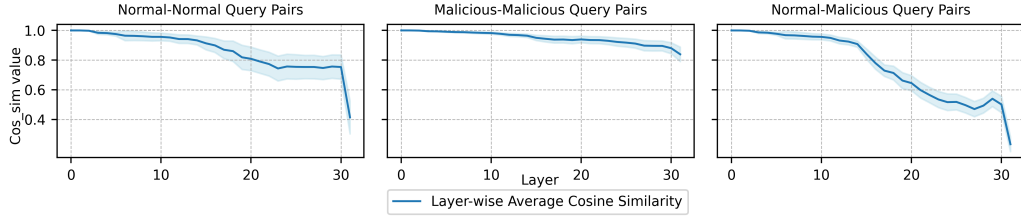


Figure 1: The layer-wise cosine similarity analysis in Phi-3-mini-4k-instruct when exposed to Normal-normal, Malicious-malicious, and Normal-malicious query pairs during inference. The shaded region represents the fluctuation range of the cosine similarity list L_C at each analysis setting, which arises from the r times random selection of different semantic query pairs. The solid lines are the numerical curve for each stratum after averaging the r sets of cosine similarity data. Statistical calculations were performed with the settings $P = 100$, $Q = 100$ and $r = 500$.

From the figure, it is evident that, the variance of layer-wise cosine similarity is fluctuated within a certain range, therefore, with $r = 500$, the mean cosine similarity curve for each layer can effectively represent the processing trend of the LLM when handling different Normal-normal, Malicious-malicious, and Normal-malicious query pairs during inference. The distinct data trends observed across different types of query pairs are intrinsic to the LLM itself.

Meanwhile, focusing solely on the trend of individual mean curves in the figure reveals the following observations: (i) Different pairs of queries within the Malicious-Malicious query class do not exhibit significant processing distinctions among the layers during the inference phase of the model. This phenomenon aligns with the behavior of aligned LLMs, which consistently output uniform rejection beginnings when refusing to answer queries (refer to Appendix A.3.3 for more details on this behavior). (ii) The differences in LLM layers are most pronounced when distinguishing questions from two distinct classes (Normal & Malicious), exceeding the distinctions observed between two semantically different normal questions. The layer-wise analysis of the other aligned LLMs shows the same trend as Figure 1, see Appendix A.3.1 for details.

When comparing the layer-wise cosine similarity trends between Normal-Normal (N-N) and Normal-Malicious (N-M) query pairs, a notable change is observed in the N-M results (illustrated in the figure on far right). This indicates the presence of safety layers. In the subsequent sections, we will conduct further analysis to confirm their existence.